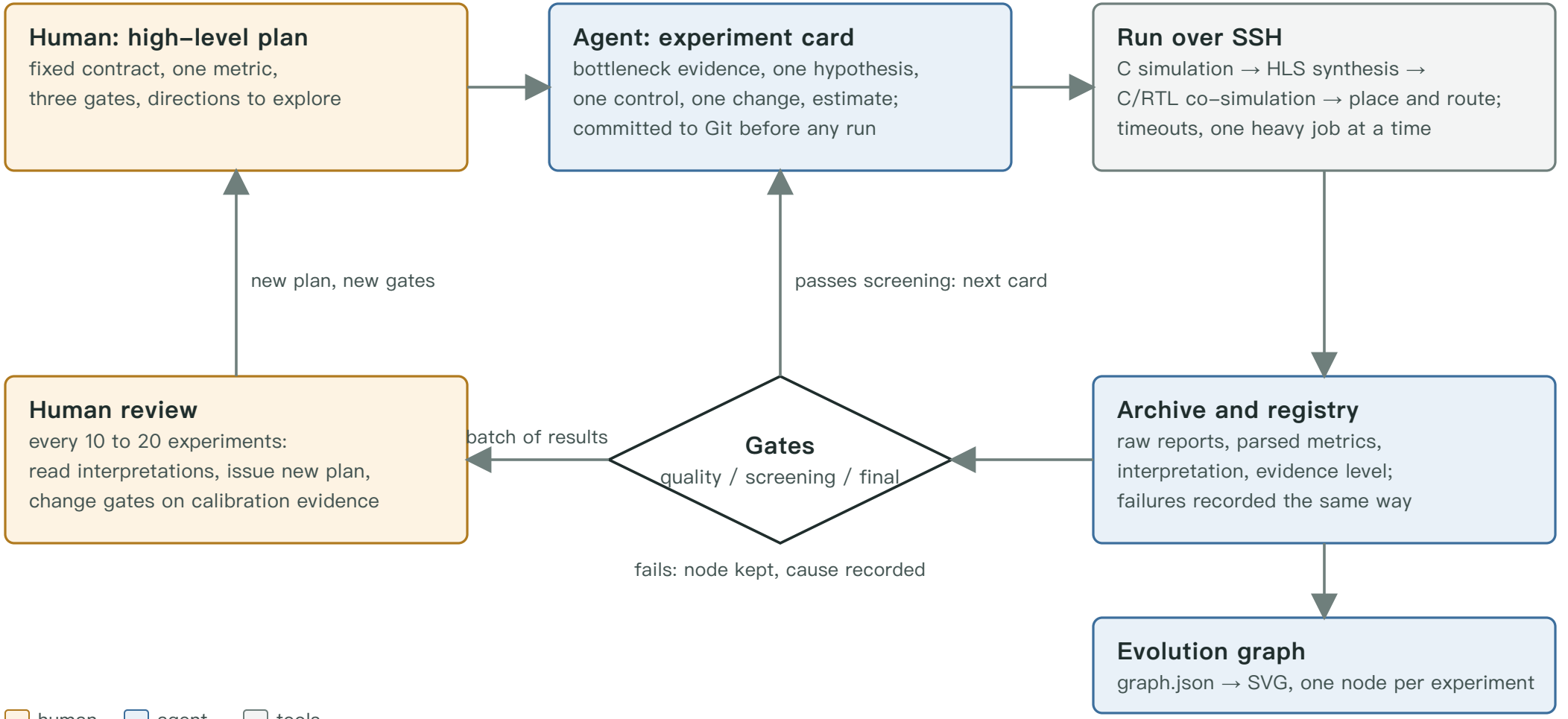
Human: high–level plan
fixed contract, one metric,
three gates, directions to explore
Agent: experiment card
bottleneck evidence, one hypothesis,
one control, one change, estimate;
committed to Git before any run
Run over SSH
C simulation → HLS synthesis →
C/RTL co–simulation → place and route;
timeouts, one heavy job at a time
new plan, new gates
passes screening: next card
Human review
every 10 to 20 experiments:
read interpretations, issue new plan,
change gates on calibration evidence
batch of results
Gates
quality / screening / final
Archive and registry
raw reports, parsed metrics,
interpretation, evidence level;
failures recorded the same way
fails: node kept, cause recorded
Evolution graph
graph.json → SVG, one node per experiment
human
agent
tools